

Markdowns under Factor-Augmenting Technology: Evidence from Indian Manufacturing, 2012–2022*

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Abstract

[Rubens et al. \(2025a\)](#) prove that the production approach to wage markdowns and factor-augmenting productivity are *not separately identified*: the same cost-share variation that a Hicks-neutral estimator reads as labour market power, a factor-biased estimator reads as labour-augmenting technology. Their remedy is to identify the inverse labour-supply elasticity outside the production function. We implement that remedy on 290,834 plant-year observations from India’s Annual Survey of Industries, 2012–2022, using a logit labour-supply model identified by firm export exposure interacted with common exchange-rate and tariff shocks, and a newly assembled state-year minimum wage series parsed from the Labour Bureau’s annual reports. We report three findings. First, the median wage markdown is **53%** ($\psi^l = 2.12$), whereas the Hicks-neutral estimates of working paper 1 put the median at $\psi_{CD} = \mathbf{0.83}$ — *below* one, and therefore implying that the median plant pays *above* marginal revenue product — and the two measures are *uncorrelated across plants* (Pearson 0.015). Second, we prove and verify numerically that under the natural CES normalisation the first-order condition $\ln(\theta^l/\theta^m) = \ln(\alpha^l\psi^l/\alpha^m)$ holds identically for every elasticity of substitution, so the markdown is invariant to the production parameters; the discrepancy with the Hicks-neutral estimate loads on labour-augmenting productivity with a coefficient exactly equal to $-k = (1 - \sigma)/\sigma$, recovered to 2×10^{-15} . Third, the direction of the bias **reverses** the Chinese

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finding: where [Rubens et al.](#) find the Hicks-neutral estimator roughly doubles the true markdown, in India it is a factor 0.37 of it. We are explicit about what the Indian data cannot deliver: with 33 states and eight usable years of minimum wage variation, σ is only weakly identified, and the cross-sectional ordering of ψ^l under a single wage coefficient is the inverse wage ordering by construction. Neither limitation touches the markdown result, and we show why.

JEL: L11, J42, O33, D24. **Keywords:** monopsony, wage markdowns, factor-augmenting technology, production function estimation, Indian manufacturing.

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1 Introduction

A decade of empirical work has concluded that employers in developing-country manufacturing pay workers less than their marginal revenue product. The dominant tool for reaching that conclusion is the *production approach*: estimate the output elasticity of labour θ^l , compare it with labour’s revenue share α^l , and read the gap — net of the product-market markup — as a wage markdown. Working paper 1 of this series applied exactly that logic to Indian manufacturing.

Rubens et al. (2025a) show the logic has a hole in it. Every production-approach markdown estimator in use assumes that productivity differences across plants are *Hicks-neutral*: they scale output without changing the ratio in which inputs are combined. Drop that assumption — allow productivity to augment labour specifically — and the output elasticity of labour is no longer a technological constant to be estimated once per industry. It becomes a plant-specific object that moves with the plant’s own labour-augmenting productivity. The cost-share variation that the Hicks-neutral estimator attributes to market power is then observationally equivalent to variation in technology. Formally: the production first-order conditions contain two unknowns per plant and supply one equation. Markdowns and factor-augmenting productivity are **not separately identified**.

This is not a hypothetical concern for the present series. Working paper 4 estimated the Demirer (2025) factor-augmenting estimator on the same Indian panel and found substantial labour-augmenting heterogeneity — while imposing $\psi^l \equiv 1$, i.e. competitive labour markets. Working paper 1 found substantial markdowns while imposing Hicks neutrality. Rubens et al.’s result says these two papers used *the same variation* and simply gave it two different names. At most one of them can be right, and nothing in either paper’s data can say which.

1.1 What this paper does

Rubens et al.’s remedy is to stop asking the production data to do impossible work and to identify the inverse labour-supply elasticity ψ^l from *labour supply* instead. If workers’ allocation across employers responds to wages in a way that can be estimated — from a discrete-choice model of where workers work — then ψ^l is pinned down without touching the production function. With ψ^l known, the production first-order condition has one unknown again, and labour-augmenting productivity A_{ft} can be recovered.

We implement this on India’s Annual Survey of Industries (ASI), 2012–2022: a plant panel of 423,127 plant-year observations covering registered manufacturing across 36 states and 24 two-digit divisions. The implementation required three pieces of data work that we regard as contributions in their own right:

1. **A labour-supply instrument.** Following Rubens et al. we need a shifter of the

wage that is orthogonal to unobserved amenities. We recover each plant’s *export share* from the raw ASI Block A fixed-width records — a variable absent from the analysis files — and interact it with the common real effective exchange rate and with industry output tariffs, in the shift-share tradition. The recovery is layout-driven: field widths are read from each wave’s own record-layout workbook and asserted against the observed record length, and all eleven waves reconcile exactly.

2. **A minimum wage series.** Rubens et al. instrument the production-side moment with county minimum wages. India has no machine-readable state-year series for our window. We parse the Labour Bureau’s annual *Report on the Working of the Minimum Wages Act, 1948*, Table 4, obtaining 321 state-years for 2012–2020, with every row validated against the report’s own internal arithmetic and carried-forward observations flagged. Reports for 2021 and 2022 do not exist and no value is extrapolated into them.
3. **An outside option.** The labour-supply model needs the population not working in registered manufacturing. We build working-age population by state and year from the National Commission on Population’s projections, with no missing cells.

1.2 Three findings

First, the markdown is large, and it is not the one working paper 1 found. The labour-supply estimate puts the median wage markdown at 53% ($\psi^l = 2.12$; weighted, 55%). The Hicks-neutral estimate on the same plants has a median ψ^l of 0.83 — *below* one, which the production approach would read as workers being paid *above* marginal revenue product at the median plant. More striking than the level gap is that the two series are essentially unrelated across plants: the plant-level correlation is 0.015 (Spearman 0.035). They are not noisy measurements of one object. They are different objects.

Second, the markdown does not depend on the production parameters at all, and we prove it. Section 4.4 shows that labour-augmenting productivity is identified only relative to the materials coefficient β^m , and that once this is imposed — rather than a spurious extra parameter being “estimated” — β^m multiplies both the labour and the materials terms of the CES aggregator and factors out of it entirely. The consequence is that $\ln(\theta^l/\theta^m) = \ln(\alpha^l\psi^l/\alpha^m)$ holds *identically, for every* σ . We verify this numerically across the whole panel at four values of σ ; the maximum absolute deviation is 5×10^{-15} . This matters because it means the paper’s headline is immune to the one parameter our data struggle to pin down.

Third, the bias runs the other way from China. The gap between the two markdown measures is exactly the gap between their elasticity ratios, and that gap loads on labour-augmenting productivity with coefficient exactly $-k = (1 - \sigma)/\sigma$ — which we

recover from a regression to fifteen decimal places, a second unit test on the algebra. The median log-gap is -0.99 : the Hicks-neutral markdown is a factor 0.37 of the true one. [Rubens et al.](#) find the reverse in Chinese nonferrous metals, where the Hicks-neutral estimator reports a 55% markdown against a true 27%. The sign of the bias is not a property of the estimator; it is a property of the covariance between labour-augmenting productivity and the labour cost share, and that covariance differs between Chinese nonferrous metals and Indian registered manufacturing. Section 11.4 takes this seriously rather than treating it as a curiosity.

1.3 What we do not claim

We are deliberate about the limits, because two of them are severe enough that a less careful reading of our own tables would produce false findings.

The elasticity of substitution is weakly identified. After the state and year effects that the overidentification test *forces* on us (Section 8.1), the two minimum-wage instruments carry nearly the same information — they correlate at 0.56 — and the model is only nominally overidentified. The weak-identification-robust confidence set for σ is unbounded on $[0.05, 6]$. We report $\hat{\sigma} = 2.08$ as a point estimate and then report every downstream object over a grid of σ , because that is what the data support. [Rubens et al.](#)’s own σ carries a standard error of 0.215 on a point estimate of 0.397; this parameter is hard everywhere.

The cross-sectional and time-series patterns in ψ^l are largely mechanical. Under a logit labour-supply model with a single wage coefficient γ , $\psi^l = 1 + 1/(\gamma W(1 - S))$ is a decreasing function of the plant’s own wage. Any ranking of plants, states, industries or ownership types by ψ^l therefore reproduces the inverse wage ranking almost exactly — in our data, at a Spearman rank correlation of -1.000 (Table 12). The same mechanism drives the apparent decline in markdowns over the decade. [Rubens et al.](#) allow a time-varying γ_t ; our instruments are too weak to support one. We therefore do *not* report the ordering of ψ^l across groups as a finding. What we do report is the level of the markdown and the log-gap against the Hicks-neutral estimate, which is built from cost shares and Cobb-Douglas elasticities and does not inherit the mechanism.

Geography is coarse. [Rubens et al.](#) define labour markets at the Chinese county level. The released ASI records contain no district identifier in any wave — the only sub-state field is the NSSO regional-office code, an administrative unit with no published geographic mapping — so our markets are states. Indian states are large, and a state-level market is a weaker approximation to a commuting zone than a county is.

1.4 Roadmap

Section 2 situates the paper. Section 3 states the identification problem formally. Section 4 develops the model, the normalisation, and the estimator, and Section 5 documents how it is computed. Section 6 describes the panel and the three data-construction exercises. Section 7 presents the labour-supply stage, Section 8 the input-ratio stage, Section 9 the production stage, and Section 10 validates the estimator on simulated data and on two identities checked at run time. Section 11 presents results; Section 12 the robustness and the honest accounting of what is weakly identified; Section 13 implications for the Indian market-power literature; Section 14 concludes. Appendices carry the derivations, the unit tests, and the data recovery details.

2 Related literature

The production approach and its assumptions. The modern markup literature descends from Hall (1988) and was made operational by De Loecker and Warzynski (2012), who showed that with one flexible input the markup is the ratio of that input’s output elasticity to its revenue share. De Loecker et al. (2020) extended the logic to input markets: where a factor is supplied imperfectly elastically, the same ratio identifies a markdown. Production-function estimation itself rests on Olley and Pakes (1996), Levinsohn and Petrin (2003), and Akerberg et al. (2015). The Indian applications include Sivadasan (2009) and, in this series, working paper 1.

Two strands have questioned what the approach measures. One concerns prices: De Loecker et al. (2016) and Bond et al. (2021) show that revenue-based elasticities need not identify markups when output prices are unobserved. The other, which is our subject, concerns *functional form*. Doraszelski and Jaumandreu (2018) and Demirer (2025) allow factor-augmenting productivity heterogeneity and show it changes elasticity estimates materially, but both must assume competitive factor markets to identify it.

The impossibility result. Rubens et al. (2025a) close the circle. They show that the Hicks-neutral markdown estimator and the factor-augmenting estimator use the same moment, and that neither the markdown nor the augmentation is identified without an outside restriction. Their solution — a labour-supply model estimated on allocation data — is the one we implement. This paper is, to our knowledge, the first independent application of their estimator outside their own Chinese setting, and the first to a large developing-country manufacturing census.

Labour-supply models of monopsony. The discrete-choice route to firm-level labour supply is due to Card et al. (2018) and Berger et al. (2022), with the inversion technique from Berry (1994). Our specification is theirs, simplified to plain logit because the Indian

data reject the nest (Section 7) — as, we note, do [Rubens et al.](#)’s, whose nesting parameter is -0.019 and statistically indistinguishable from zero.

Why functional form binds harder in developing-country data. The Hicks-neutrality assumption is not equally costly everywhere. It is a restriction on how much plants differ in *how* they combine inputs, not in how much output they get from a given bundle. In a mature industry where technique is close to uniform, the restriction is nearly free. Indian registered manufacturing is the opposite case: within a single two-digit division it contains plants that are essentially manual and plants that are close to the international frontier, and the dispersion of measured productivity in Indian manufacturing has been documented as extreme ([Hsieh and Klenow, 2009](#)). The evidence in working paper 4 that labour-augmenting heterogeneity is large in exactly this panel is the reason the present exercise is worth doing here rather than as a methodological footnote.

Monopsony in low-income labour markets. A parallel literature argues that employer wage-setting power is pervasive where workers are immobile, information is poor and outside options are thin ([Brooks et al., 2021](#); [Amodio and de Roux, 2022](#)). That literature mostly identifies markdowns from *labour supply*, as we do, rather than from production. Our contribution to it is to put the two identification strategies on the same plants and to show what happens: they agree that markdowns are large and they disagree completely about which plants have them. That disagreement is not a puzzle to be split; [Rubens et al.](#)’s result says the production estimate is the one without a claim to identification once technology is factor-biased, and our decomposition shows exactly what it is picking up instead.

Weak identification. Our inference leans on [Anderson and Rubin \(1949\)](#) and [Stock and Wright \(2000\)](#) more than is usual in this literature, because both of our identifying designs are on the weak side and we would rather say so with the right tools than report a conventional standard error that assumes the problem away. The practical guidance we follow is [Andrews et al. \(2019\)](#): report the identification-robust set, and report it *instead of* the conventional interval where the two disagree.

Monopsony and minimum wages in India. [Soundararajan \(2019\)](#) is the closest antecedent: it reads Indian minimum wages through an explicitly monopsonistic lens using state-time variation in statutory rates. [Menon and Rodgers \(2017\)](#) and [Mansoor and O’Neill \(2021\)](#) construct minimum wage databases for India; the latter, the most complete, covers 1999–2011 and therefore stops before our window opens, which is why we parse the Labour Bureau reports directly (Appendix C). Evidence on Indian labour market power more broadly includes [Hasan et al. \(2007\)](#) on labour regulation and wages.

3 The identification problem, stated formally

3.1 Setup

A plant f in year t produces gross output Q_{ft} from labour L_{ft} , materials M_{ft} and capital K_{ft} . Productivity has two components: a Hicks-neutral term Ω_{ft} that scales output, and a labour-augmenting term A_{ft} that multiplies labour inside the production function:

$$Q_{ft} = G(A_{ft}L_{ft}, M_{ft}, K_{ft}; \beta) \cdot \Omega_{ft}. \quad (1)$$

Write θ_{ft}^l and θ_{ft}^m for the output elasticities of labour and materials, and $\alpha_{ft}^l = W_{ft}^l L_{ft} / (P_{ft} Q_{ft})$, $\alpha_{ft}^m = W_{ft}^m M_{ft} / (P_{ft} Q_{ft})$ for their revenue shares. Let ψ_{ft}^j denote the inverse supply elasticity of input j , so that $\psi^j = 1$ under competitive supply and $\psi^j > 1$ when the plant faces an upward-sloping supply curve.

Cost minimisation delivers the two identities the whole literature runs on (De Loecker and Warzynski, 2012; De Loecker et al., 2020):

$$\mu_{ft}^p = \frac{\theta_{ft}^m}{\alpha_{ft}^m \psi_{ft}^m}, \quad (2)$$

$$\psi_{ft}^l = \frac{\theta_{ft}^l}{\theta_{ft}^m} \cdot \frac{\alpha_{ft}^m}{\alpha_{ft}^l} \cdot \psi_{ft}^m. \quad (3)$$

Throughout we impose $\psi^m = 1$: materials are bought competitively. The wage markdown — the share of marginal revenue product that the plant retains — is

$$\mu_{ft}^w = \frac{\psi_{ft}^l - 1}{\psi_{ft}^l} = \frac{MRPL_{ft} - W_{ft}^l}{MRPL_{ft}}. \quad (4)$$

3.2 Where the identification fails

Equation (3) looks like a formula for ψ^l in terms of observables. It is not, once A_{ft} varies. Under (1) the elasticities θ_{ft}^l and θ_{ft}^m are *functions of* A_{ft} : a plant with more labour-augmenting productivity has, in effective units, more labour, and therefore a different marginal product of labour relative to materials. So (3) has unknowns on both sides even after the technology parameters β are estimated.

Two literatures resolve this by assumption, in opposite directions:

- **Hicks-neutral estimators** (De Loecker and Warzynski, 2012; De Loecker et al., 2020; working paper 1) set $A_{ft} = A$ constant. Then θ^l and θ^m are technological constants within an industry, and *all* plant-level variation in α^l / α^m is read as variation in ψ^l .
- **Factor-biased estimators** (Doraszelski and Jaumandreu, 2018; Demirer, 2025;

working paper 4) set $\psi^l = 1$. Then *all* of the same variation is read as variation in A_{ft} .

Both use the same moment. Neither can test the other’s assumption. This is [Rubens et al.](#)’s result, and it is why working papers 1 and 4 of this series cannot both be taken at face value.

3.3 The way out

The restriction that breaks the tie must come from outside the production function. [Rubens et al.](#)’s proposal is a *labour-supply model*: if the allocation of workers across plants responds to wages in an estimable way, ψ^l is identified from that response. Equation (3) can then be read forwards — it delivers θ^l/θ^m , hence A_{ft} — rather than being asked to deliver two unknowns at once. The rest of this paper is the implementation of that idea.

4 The model and its identification

4.1 Labour supply

Worker j chooses among plants in a local labour market ℓ (a state-year), with the outside option of not working in registered manufacturing. Utility from plant f is

$$U_{jft} = \gamma_t W_{ft}^l + \xi_{ft} + \varepsilon_{jft}, \quad (5)$$

with ξ_{ft} a plant amenity and ε i.i.d. extreme value. The [Berry \(1994\)](#) inversion turns the resulting share equation into a regression:

$$\ln S_{ft} - \ln S_{0t} = \gamma W_{ft}^l + \text{controls} + \text{FE} + \xi_{ft}, \quad (6)$$

where S_{ft} is plant f ’s share of the market’s working-age population and S_{0t} the share choosing the outside option. The implied inverse labour-supply elasticity is

$$\boxed{\psi_{ft}^l = 1 + \frac{1}{\gamma W_{ft}^l (1 - S_{ft})}} \quad (7)$$

[Rubens et al.](#) use the nested version, with plants nested in county-industry cells. We tested the nest and it is rejected at its boundary in the Indian data (Section 7), exactly as it is in theirs, so (6) is the plain logit.

Two properties of (7) deserve emphasis now rather than in a footnote later, because they discipline what we are allowed to conclude. First, $\psi^l > 1$ for any $\gamma > 0$: **the sign of the markdown is mechanical and carries no information**. Second, ψ^l is decreasing

in the plant's own wage, so cross-sectional orderings of ψ^l are inverse wage orderings. Both are properties of the model, not findings from the data.

4.2 Production: CES with labour-augmenting technology

We take [Rubens et al.](#)'s functional form. With elasticity of substitution σ , returns to scale ν , and $k \equiv (\sigma - 1)/\sigma$:

$$Q_{ft} = \left[(A_{ft}L_{ft})^k + \beta^m M_{ft}^k + \beta^k K_{ft}^k \right]^{\nu/k} \Omega_{ft}. \quad (8)$$

Both unobservables follow AR(1) laws of motion,

$$\tilde{\omega}_{ft} = \rho^\omega \tilde{\omega}_{f,t-1} + v_{ft}^\omega, \quad (1 - \sigma)a_{ft} = \rho^a (1 - \sigma)a_{f,t-1} + v_{ft}^a, \quad (9)$$

with lower case denoting logs. Note that the state variable in the second equation is $(1 - \sigma)a$, not a ; this is [Rubens et al.](#)'s parameterisation and it turns out to be the convenient one.

4.3 The input-ratio equation

Combining the CES form of θ^l/θ^m with the first-order condition (3) and taking logs gives, after the algebra in Appendix A, a linear relation. Define the observables

$$x_{ft} \equiv \ln \frac{\alpha_{ft}^l \psi_{ft}^l}{\alpha_{ft}^m}, \quad u_{ft} \equiv m_{ft} - l_{ft}, \quad v_{ft} \equiv u_{ft} + x_{ft}. \quad (10)$$

A short calculation shows $v_{ft} = w_{ft}^l + \ln \psi_{ft}^l - w_{ft}^m$: it is the *markdown-adjusted relative input price*, the true marginal cost of labour relative to materials. The estimating equation is then

$$\boxed{u_{ft} = c + \sigma v_{ft} + (1 - \sigma)a_{ft}} \quad (11)$$

— a demand-for-inputs relation whose slope is the elasticity of substitution and whose error is the labour-augmenting state. Writing $G_{ft}(\sigma) = u_{ft} - \sigma v_{ft}$, the AR(1) in (9) makes the innovation

$$v_{ft}^a = u_{ft} - \rho^a u_{f,t-1} - \sigma v_{ft} + \sigma \rho^a v_{f,t-1}, \quad (12)$$

which is **bilinear** in the two parameters. This is exploited hard in estimation: for any fixed σ the moment is linear in ρ^a , so ρ^a is concentrated out in closed form and the profiled objective is a clean one-dimensional curve in σ — which is simultaneously the estimator and, evaluated pointwise, the weak-identification-robust [Stock and Wright \(2000\)](#) statistic. Stage 3 has exactly the same bilinear shape in $(\nu/k, \rho^\omega)$, so a single routine serves both.

The identifying moments are [Rubens et al.](#)'s:

$$\mathbb{E}[v_{ft}^a \mid w_{f,t-1}^l, w_{\ell t}^{\min}, w_{\ell,t-1}^{\min}] = 0. \quad (13)$$

The lagged own wage is predetermined with respect to this period's technology innovation; the statutory minimum wage is a policy shifter of labour cost that no individual plant's technology shock causes.

4.4 The normalisation, and why it is the key to the paper

A is identified only relative to β^m : scaling A up and β^m down leaves (8) unchanged. Rather than pretend to estimate both, we impose the restriction and follow it through, which turns out to have a striking consequence.

Write $a_{ft} = \tilde{a}_{ft} + \kappa$ where $\tilde{a}_{ft} = G_{ft}(\sigma)/(1 - \sigma)$ is observable given σ , and $\kappa = -\sigma \ln \beta^m / (1 - \sigma)$ absorbs the normalisation. The labour term of the aggregator becomes $e^{k\kappa} \exp(k(\tilde{a}_{ft} + l_{ft}))$, and

$$k\kappa = \frac{\sigma - 1}{\sigma} \cdot \frac{-\sigma \ln \beta^m}{1 - \sigma} = \ln \beta^m \quad \text{exactly.} \quad (14)$$

So β^m multiplies *both* the labour and the materials term, and factors out of the bracket into the intercept of (8). Only the ratio $b \equiv \beta^k / \beta^m$ survives:

$$\boxed{\tilde{D}_{ft} = \exp(k(\tilde{a}_{ft} + l_{ft})) + M_{ft}^k + b K_{ft}^k, \quad q_{ft} = \frac{\nu}{k} \ln \tilde{D}_{ft} + \text{const} + \omega_{ft}.} \quad (15)$$

Now compute the elasticity ratio this implies. Using $k\tilde{a}_{ft} = -G_{ft}(\sigma)/\sigma = -u_{ft}/\sigma + v_{ft}$ and $l - m = -u$:

$$\ln \frac{\theta_{ft}^l}{\theta_{ft}^m} = k(\tilde{a}_{ft} + l_{ft}) - k m_{ft} = -\frac{u_{ft}}{\sigma} + v_{ft} + \left(\frac{\sigma - 1}{\sigma} \right) (-u_{ft}) = v_{ft} - u_{ft} = x_{ft}. \quad (16)$$

That is, $\ln(\theta^l/\theta^m) = \ln(\alpha^l \psi^l / \alpha^m)$ **identically, for every value of σ** . The first-order condition is satisfied by construction rather than imposed as a restriction, and:

The wage markdown ψ^l , the markdown rate μ^w , and every comparison between them and the Hicks-neutral estimates are invariant to the production parameters. Only A_{ft} itself, the individual elasticities $\theta^l, \theta^m, \theta^k$, the returns to scale ν , and hence the product markup μ^p depend on σ .

We verify (16) numerically over the whole estimation panel at four values of σ ; the largest absolute deviation is 4.8×10^{-15} (Appendix B). This is the reason the weak identification of σ reported in Section 8 is an inconvenience rather than a threat.

4.5 The production stage

With $\tilde{a}_{ft}$ in hand, (15) is estimated for (b, ν, ρ^ω) on [Akerberg et al. \(2015\)](#) timing moments,

$$\mathbb{E}[v_{ft}^\omega \mid l_{f,t-1}, m_{f,t-1}, k_{ft}, k_{f,t-1}] = 0, \quad (17)$$

capital entering contemporaneously because it is chosen a period in advance. The output elasticities follow directly:

$$\theta_{ft}^l = \nu \frac{\exp(k(\tilde{a}_{ft} + l_{ft}))}{\tilde{D}_{ft}}, \quad \theta_{ft}^m = \nu \frac{M_{ft}^k}{\tilde{D}_{ft}}, \quad \theta_{ft}^k = \nu \frac{b K_{ft}^k}{\tilde{D}_{ft}}, \quad (18)$$

summing to ν , with the markup from (2).

One control we cannot include. [Rubens et al.](#) carry a firm output price control $\beta^p p_{ft}$ in ω . The ASI has no plant-level output price; output is deflated by a two-digit wholesale price index, so plant-level price deviations remain in ω_{ft} . This is the [De Loecker et al. \(2016\)](#) omitted-price problem and we do not solve it. We report industry-by-year effects as the closest available substitute and flag the limitation where it bites.

5 Implementation

There is no Stata on the machine this project runs on, so the whole estimator is written in Python over NumPy and SciPy. This section documents it completely, because several of the choices are not cosmetic.

5.1 One routine for two stages

The residual of the input-ratio stage, (12), and the residual of the production stage are the same algebraic object. Writing the production-stage residual out with $\Phi_{ft} = \ln \tilde{D}_{ft}$ and $s = \nu/k$,

$$v_{ft}^\omega = q_{ft} - \rho^\omega q_{f,t-1} - s \Phi_{ft} + s \rho^\omega \Phi_{f,t-1},$$

which is (12) with (u, v) replaced by (q, Φ) and (σ, ρ^a) by (s, ρ^ω) . Both are therefore estimated by a single routine that takes four demeaned vectors, an instrument matrix, weights and cluster identifiers.

The routine exploits the bilinearity. For a fixed value of the first parameter the moment vector is *linear* in the second,

$$\bar{g}(s, \rho) = A - \rho B - sC + s\rho E, \quad A = Z'W u_t, \quad B = Z'W u_{t-1}, \quad C = Z'W v_t, \quad E = Z'W v_{t-1},$$

so the GMM criterion is a quadratic in ρ and is minimised in closed form: $\rho^*(s) =$

$(b'\hat{W}a)/(b'\hat{W}b)$ with $a = A - sC$, $b = B - sE$. The estimator is then a one-dimensional sweep over s . Two consequences follow, and both are used:

1. There is no multi-dimensional optimiser and therefore no starting-value or local-minimum problem. Earlier work in this series found three-dimensional Nelder–Mead on this class of objective unstable; concentrating is both faster and deterministic.
2. The profiled criterion *is* the [Stock and Wright \(2000\)](#) statistic for s with ρ concentrated out. Inverting it gives a confidence set valid under weak identification at no additional cost — which is exactly what Section 8 needs.

Weighting is two-step: a first pass with $(Z'WZ)^{-1}$, then the cluster-robust efficient weight $\hat{W} = [\sum_c (\sum_{i \in c} w_i z_i \hat{v}_i) (\cdot)']^{-1}$ evaluated at the first-pass estimate, then a second sweep. Standard errors use the analytic Jacobian; because the residual is bilinear the Jacobian is available in closed form, $\partial v / \partial s = -(v_t - \rho v_{t-1})$ and $\partial v / \partial \rho = -(u_{t-1} - s v_{t-1})$.

5.2 Fixed effects

All specifications absorb high-dimensional fixed effects — plant effects in Stage 1 run to 65,001 levels — by weighted alternating projections rather than dummy variables, iterating group-mean subtraction to a tolerance of 10^{-9} . Because the demeaning is linear it commutes with the parameters, so in Stages 2 and 3 the four data vectors and the instruments are demeaned *once*, outside the parameter sweep. Everything inside the sweep is then $k \times k$ linear algebra on 3×3 or 4×4 matrices, which is what makes a 1,200-point profile and a 200-draw bootstrap of the full chain feasible.

5.3 The outer search in Stage 3

Φ_{ft} depends on b , so b cannot be concentrated out. It is found by a coarse grid over $\ln b \in [-20, 20]$ in steps of 0.5, followed by a refinement in steps of 0.05 around the coarse minimum; at each $\ln b$ the inner concentrated routine runs in full. The CES index is evaluated as a log-sum-exp,

$$\Phi_{ft} = \text{logsumexp}(k(\tilde{a}_{ft} + l_{ft}), k m_{ft}, \ln b + k k_{ft}),$$

with the elasticity shares from the corresponding softmax. This is not a nicety: k changes sign with σ , and the three terms differ by many orders of magnitude at plausible parameter values, so the naive expression overflows.

A sign trap is worth recording because it is easy to miss. The profiled parameter in Stage 3 is $s = \nu/k$, and $k < 0$ whenever $\sigma < 1$. A grid over positive s therefore silently searches only negative returns to scale for every σ below one. We grid over ν and map to $s = \nu/k$, which is sign-safe by construction.

5.4 Inference

Standard errors and the weight matrices cluster on the plant throughout. The bootstrap of Section 12 resamples plants with replacement and re-runs every stage, so that a resampled plant enters as a *distinct* cluster; this is what makes the fixed-effect absorption and the clustered weight matrices behave correctly under resampling. Stage-1 inference additionally reports the [Anderson and Rubin \(1949\)](#) set, because the first-stage F of about 13 is at the edge of conventional strength.

5.5 Reproducibility

Each stage is a numbered script that writes a log and one or more result CSVs. Every table in this paper is generated from those CSVs by a separate script and included by `\input`, and every figure by the same route, so **no number in any table or figure is typed by hand**. Numbers quoted in the running text are read from those same logs and CSVs, and where the text and a table state the same quantity it is the same number by construction. The two identities of Appendix B are **asserted** at run time, so a change that broke the algebra would stop the pipeline rather than produce a plausible-looking table.

6 Data

6.1 The plant panel

The Annual Survey of Industries is India’s census of registered manufacturing: a complete enumeration of larger factories and a state-stratified sample of smaller ones, with a multiplier that grosses the sample to the population. We use the Common-Factory-Id panel files for 2012–13 through 2022–23, which carry a stable plant identifier across waves. All monetary values are deflated to 2011–12 prices with two-digit wholesale price indices (output), the all-commodity index (materials) and the machinery index (capital). Sampling weights are retained throughout: they enter the estimation as weights and the descriptive statistics as population weights.

After the standard screens the panel is **423,127 plant-years** across 36 states and 24 two-digit divisions, defining 377 state-year labour markets. The median plant employs 89 persons and pays Rs 182,802 per worker per year. Labour is total persons engaged; the wage is total emoluments (wages, bonus, provident fund and welfare) per person, so the markdown is measured against total labour compensation rather than the wage line alone.

Table 1: Descriptive statistics, estimation panel, weighted by the ASI multiplier. Monetary values are in 2011–12 prices. The revenue-share *means* are dominated by a small number of plants reporting costs far in excess of recorded revenue and are shown only for completeness; the quantiles are the informative summary, and the estimator uses ratios in logs.

	p25	Median	p75	Mean
Gross output, real (Rs lakh)	168.475	1,065.383	5,148.035	10,687.450
Materials, real (Rs lakh)	119.030	772.117	3,735.821	7,871.255
Net fixed capital, real (Rs lakh)	36.258	211.329	1,202.491	4,462.104
Persons engaged	16.000	62.000	209.000	233.326
Emoluments per person (Rs)	113,297.723	182,923.800	286,369.899	228,224.304
Labour revenue share α^l	0.049	0.098	0.186	5.762
Materials revenue share α^m	0.611	0.755	0.866	11.016

Table 2: Sample construction. Each screen is re-applied in order by the results script rather than copied from a log, so the ladder cannot drift out of step with the estimation samples. The three stages then impose their own data requirements, listed below the table.

Screen	Retained	Dropped	Plants
ASI plant panel, 2012–2022	423,127	—	—
wage trimmed at 0.5 / 99.5 percentiles	418,895	4,232	84,136
market screen: mfg employment / working-age pop ≥ 0.10	413,438	5,457	82,903
positive employment	413,438	0	82,903
finite input-ratio variables (u, x, v)	413,438	0	82,903
<i>Stage-specific requirements</i>			
Stage 1 (labour supply): a lagged export share	290,834		65,001
Stage 3 (production): consecutive years	289,760		64,802
Stage 2 (input ratio): minimum wage at t and $t - 1$	223,242		56,907

The Stage-2 window is 2013–2020 rather than 2013–2022 because the Labour Bureau published no *Report on the Working of the Minimum Wages Act* for 2021 or 2022. No value is extrapolated into those years.

Sample screens. Two screens are applied and both are consequential enough to state. Wages are trimmed at the 0.5 and 99.5 percentiles. And state-years in which recorded registered-manufacturing employment exceeds 10% of the state’s working-age population are dropped, because the logit outside option is not meaningful there: this removes Daman & Diu and Dadra & Nagar Haveli, two small union territories that host migrant-worker enclaves, amounting to 19 of 377 state-years.

6.2 Three constructed inputs

Export share, recovered from the raw records. The shift-share instrument needs plant-level trade exposure, which the analysis files do not carry. We wrote a fixed-width parser for the raw ASI Block A and Block B records across all eleven waves. The parser is *layout-driven*: field names and widths are read from each wave’s own record-layout workbook, offsets are formed by cumulative sum, and the implied record length is asserted against the observed one. All eleven waves reconcile exactly. The result is 719,496 factory-years with export share complete; 7.4% of factory-years report positive exports. Two traps encountered in the recovery are documented in Appendix D, because both would silently corrupt an ownership analysis.

Minimum wages. Appendix C documents the source and the validation. In brief: the Labour Bureau’s annual *Report on the Working of the Minimum Wages Act, 1948*, Table 4, gives the lowest and highest scheduled minimum wage in rupees per day for unskilled workers in each state. We parse nine annual reports and obtain **321 state-years for 2012–2020**. Every parsed row must satisfy the report’s own internal arithmetic (range = highest – lowest) or it is discarded and counted. States whose report footnote marks the figures as repeated from the previous year — 63 state-years, 19.6% — are flagged, because they carry no independent time variation. **Reports for 2021 and 2022 do not exist**; those years have no minimum wage and nothing is extrapolated forward. The series retains 30% of its log variation after state and year effects, against 9.8% for the industry tariff, which is why it is the better instrument for the production-side moment.

Working-age population. The outside option requires the population not employed in registered manufacturing. We take working-age (15–59) population by state and year from the National Commission on Population’s *Population Projections 2011–2036*, supplemented by Census 2011 for the smallest union territories. There are no missing state-years.

What is not available. There is no district field in any ASI wave. The only sub-state identifier is the NSSO regional/sub-regional office code, an administrative survey unit with no published mapping to geography. **The state is therefore the finest labour**

market we can define, against Rubens et al.’s counties. We regard this as the most important data limitation of the paper.

7 Stage 1: labour supply

7.1 Specification

The market is the state-year; shares are computed against working-age population, with the numerator each plant’s own employment and the denominator the population total, so that weights enter the denominator but not the numerator. We first estimated the nested logit of Card et al. (2018), nesting plants within state-industry cells. The nesting parameter is -0.114 at the two-digit nest, -0.010 at the five-digit nest and $+0.045$ with industry-only nests: **the nest is rejected at its theoretical boundary** $\varsigma \in [0, 1)$, and γ is invariant to the nest definition. We therefore impose $\varsigma = 0$ and estimate the plain logit. We note that Rubens et al. find the same thing — a nesting parameter of -0.019 , insignificant — and read it as industries and localities being close to symmetric substitutes for workers.

7.2 Instruments and the fixed-effect problem

Wages are endogenous to amenities, so (6) needs instruments. Following Rubens et al. we use exposure to common shocks: the plant’s *lagged* export share interacted with the real effective exchange rate, and with the industry output tariff. Lagging makes exposure predetermined with respect to the current amenity shock; the main effect of exposure is controlled, so identification is from differential exposure to common shocks in the shift-share tradition.

Table 3 shows that the fixed-effect choice, not the instrument choice, is what matters. Without plant fixed effects γ changes sign across instrument sets (-1.20 , $+0.12$, $+0.13$) and the Hansen test rejects ($p = 0.010$ and 0.033). That is the signature of a persistent plant-level amenity sitting in ξ_{ft} and correlated with export status — exporters are different employers in ways that persist. Adding plant fixed effects removes it: γ is stable at about 0.43 across every instrument set and both fixed-effect variants, and the Hansen test passes comfortably ($p = 0.75$ to 0.94).

There is *no* strength-for-consistency trade-off here, and we correct an earlier version of our own record that claimed one. First-stage F ranges 5.9 – 20.3 without plant effects and 5.3 – 22.4 with them — comparable, and *higher* with plant effects for the exposure $\times$ exchange-rate instrument alone (22.4 against 5.9). The larger figures first reported came from a first-stage statistic that wrongly included the *included* exogenous control in the Wald test; correcting it lowered every F computed without plant effects and left the

preferred specification at $F = 13.0$. The instruments are therefore moderately strong *everywhere*, which is why the weak-identification-robust set below carries the inference.

Table 3: Stage-1 labour supply. Firm fixed effects are the binding specification choice: without them γ changes sign across instrument sets and the overidentification test rejects, which is what a persistent firm-level amenity correlated with export status looks like. With firm effects γ is stable and the test passes. Instrument strength is comparable either way (F 5.9–20.3 without firm effects, 5.3–22.4 with), so there is no trade-off — but it is only moderate throughout, which is why the Anderson–Rubin set carries the inference in the text.

Fixed effects	Instruments	γ	(s.e.)	F	Hansen p	median ψ^l
state+NIC2+year	exp x REER	-1.1955	(0.7891)	5.9	—	—
state+NIC2+year	exp x REER + exp x tar	0.1174	(0.1985)	20.3	0.010	5.24
state+NIC2+year	+ tariff	0.1348	(0.1838)	15.4	0.033	4.69
FIRM+year	exp x REER	0.4255	(0.1612)	22.4	—	2.17
FIRM+year	exp x REER + exp x tar	0.4441	(0.1430)	13.0	0.749	2.12
FIRM+year	+ tariff	0.4334	(0.1370)	9.2	0.917	2.15
FIRM+state x year	exp x REER	0.4294	(0.2168)	12.3	—	2.16
FIRM+state x year	exp x REER + exp x tar	0.4369	(0.1793)	7.9	0.938	2.14
FIRM+state x year	+ tariff	0.4309	(0.1780)	5.3	0.927	2.15

7.3 Result

The preferred specification — plant and year effects, instrumented by exposure $\times$ exchange rate and exposure $\times$ tariff — gives

$$\hat{\gamma} = 0.4441 \text{ (0.1430)}, \quad t = 3.11, \quad N = 290,834, \quad 65,001 \text{ plant clusters},$$

with a first-stage F of 13.0 and a Hansen statistic of 0.103 on one degree of freedom ($p = 0.749$). Because $F \approx 13$ sits at the edge of conventional strength, we report the [Anderson and Rubin \(1949\)](#) weak-identification-robust confidence set, which is valid regardless of instrument strength:

$$\gamma \in [0.195, 1.030] \text{ at } 95\%.$$

It excludes zero, so the wage coefficient is signed; but the *level* of ψ^l is only loosely pinned, and we carry that through everywhere. At $\hat{\gamma}$ the implied markdowns are

$$\psi^l : \text{p10 } 1.48, \text{ p25 } 1.72, \text{ median } 2.12, \text{ p75 } 2.80, \text{ p90 } 4.10,$$

a median wage markdown of **52.8%**. Across the Anderson–Rubin set the median ψ^l ranges from 1.48 to 3.55, i.e. a markdown between **33% and 72%**. That interval, not the point estimate, is the honest statement of what the Indian data say about the level of monopsony power.

8 Stage 2: the elasticity of substitution

The input-ratio GMM of (12)–(13) is estimated on the 223,242 plant-years (56,907 plants, 33 states, 2013–2020) that have consecutive observations and a minimum wage in both years. The window is shorter than the panel because the Labour Bureau reports stop in 2020.

8.1 The fixed effects are chosen by the test

Table 4: Input-ratio GMM for the elasticity of substitution. Panel (a) varies the fixed effects; the overidentification test, not our preference, selects them. Panel (b) varies the instrument set at the selected fixed effects. The final column of panel (a) is the weak-identification-robust (Stock–Wright) 95% set for σ on the grid $[0.05, 6]$.

(a) Fixed effects						
Fixed effects	σ	(s.e.)	ρ^a	J	p	Robust set
year	3.105	(1.351)	0.996	2.81	0.094	[2.69, 4.06]
state + year	2.080	(0.173)	0.525	0.85	0.356	UNBOUNDED
state + NIC2 + year	2.320	(0.267)	1.213	1.90	0.169	UNBOUNDED
state + NIC2 x year	2.315	(0.218)	1.221	0.87	0.352	UNBOUNDED

(b) Instruments, state + year effects						
Instruments	σ	(s.e.)	ρ^a	J	d.f.	p
RWX: w(t-1), wmin(t), wmin(t-1)	2.080	(0.173)	0.525	0.85	1	0.356
+ minimum-wage bite x d.wmin	2.085	(0.127)	0.511	0.82	2	0.662
+ capital k(t) (ACF timing)	2.545	(3.131)	0.938	29.08	2	0.000
+ k(t), u(t-2), w(t-2)	1.565	(0.070)	0.954	124.45	4	0.000

Panel (a) of Table 4 reports the specification search, and we want to be explicit that the overidentification test settled it rather than our judgement. With year effects only, the model is rejected at the 10% level and σ runs towards the top of the grid. The reason is economic: **minimum wage levels are higher in richer states**, and richer states also differ permanently in labour-augmenting technology, so the AR(1) quasi-difference leaves a state component in the residual that the instrument is correlated with. Adding state effects removes it and the test passes ($J = 0.85$, $p = 0.356$). State and year effects are therefore the baseline.

Panel (b) makes the converse point. It is tempting to add moments — lagged capital, second lags of inputs and wages — because they are predetermined *if productivity is exactly AR(1)*, and because they would sharpen ρ^a enormously. They also reject the model overwhelmingly: $J = 29.1$ on adding capital, $J = 124.5$ on adding second lags, both with $p < 0.001$. We read that as evidence that the AR(1) is an approximation, and we **report those specifications and drop them** rather than keeping the precision they appear to buy. The estimator uses exactly the moment set Rubens et al. propose.

8.2 σ is weakly identified, and here is why

The point estimate is $\hat{\sigma} = 2.08$ (s.e. 0.173) with $\rho^a = 0.53$, and the model is not rejected. But the profiled objective — which is the Stock–Wright statistic and therefore a valid confidence set under weak identification — never rises above 1.83 anywhere on $[0.05, 6]$, against a 5% critical value of 5.99. **The robust 95% set for σ is unbounded on the grid.**

The diagnosis is specific. After the state and year effects the test forces on us, $\ln w_t^{\min}$ and $\ln w_{t-1}^{\min}$ are correlated at 0.558 with each other. Three moments that span barely more than two directions cannot identify two parameters and test the overidentifying restriction as well: the model is only *nominally* overidentified. The first-stage F of the two minimum-wage instruments for v is 7.5. Only 36% of the minimum wage’s log variation survives state and year effects, and what survives is smooth. With 33 states and eight usable years, this is a data limitation and not a fixable specification error. We note that Rubens et al.’s own σ has a standard error of 0.215 around a point estimate of 0.397: this parameter is difficult in their setting too.

Section 4.4 is what makes this survivable. Because the first-order condition holds identically for every σ , the markdown results are untouched. We therefore report $\hat{\sigma}$ as a point estimate and report every σ -dependent object over a grid (Table 13).

8.3 Sensitivity to Stage 1

σ inherits Stage-1 uncertainty because ψ^l enters v . Across the Anderson–Rubin set for γ : $\gamma = 0.195$ gives $\sigma = 3.36$; $\gamma = 0.444$ gives 2.08; $\gamma = 1.030$ gives 1.46. Under the competitive benchmark $\psi^l \equiv 1$ the estimate is 3.88 with a standard error of 4.59. All are within the unbounded robust set, which is the point.

9 Stage 3: the production function

Estimated on all 289,760 plant-years with consecutive observations — the minimum wage is not needed here, so 2021 and 2022 return — the pooled specification gives $\nu = 1.11$ and $\rho^\omega = 0.89$ at $\hat{\sigma}$. **But the overidentification test rejects at every σ** , with $J = 88.8$ against one degree of freedom. We did not proceed past that. Two checks, in order.

Is it power? Partly. With 289,760 observations and 64,802 clusters a one-degree-of-freedom test rejects economically trivial misspecification, so the question worth asking is whether the *parameters* move when the moments change. Dropping each moment in turn moves ν only within $[0.905, 1.155]$ and ρ^ω within $[0.792, 0.931]$. The exception is informative: dropping lagged capital k_t sends b from 0.08 to 299, which simply says that k_t is the moment identifying the capital coefficient.

Table 5: Stage-3 production parameters, pooled, across values of σ . The overidentification test rejects at every σ ; Section 9 shows this is pooling rather than the model, and reports the by-industry estimates that are used throughout. $b = \beta^k/\beta^m$ is the only capital coefficient identified after the normalisation of Section 4.4.

Fixed effects	σ	k	b	ν	(s.e.)	ρ^ω	J	p
year	0.40	-1.500	6.773e-05	0.510	(0.016)	0.995	119.65	0.000
year	0.80	-0.250	7.869e-05	0.795	(0.010)	0.903	166.27	0.000
year	1.50	0.333	1.003e-01	1.170	(0.011)	0.890	127.91	0.000
year	2.08	0.519	8.208e-02	1.110	(0.007)	0.886	88.84	0.000
year	3.00	0.667	7.065e-02	1.085	(0.005)	0.881	61.86	0.000
state + NIC2 $\times$ year	2.08	0.519	1.165e-01	1.090	(0.006)	0.865	61.95	0.000

By industry (23 NIC-2 divisions, σ pooled): ν has median 1.085 (p25 1.040, p75 1.105) and ρ^ω median 0.845. The overidentification test rejects at the 5% level in 3 of 23 divisions (median $p = 0.714$).

Is it pooling? Mostly, yes — and this is the resolution. Imposing a single (ν, b) on all of manufacturing is a strong restriction that working paper 1 does not impose on its Cobb-Douglas. Estimating Stage 3 *separately by two-digit division*, holding σ pooled because Stage 2 cannot support two dozen weakly-identified σ s, the test **rejects in only 3 of 23 divisions** (median $p = 0.714$), with ν median 1.085 and ρ^ω median 0.845. The by-industry estimates are therefore the preferred ones and are used for every elasticity reported below. This is also the like-for-like comparison with working paper 1, whose Cobb-Douglas is estimated industry by industry. For reference, [Rubens et al.](#) report ν between 0.983 and 1.046 and ρ^ω near 0.98.

10 Validation

10.1 Monte Carlo

Before taking the estimator to the ASI we reproduced the Monte Carlo design of [Rubens et al. \(2025b\)](#): 50 markets of 5 plants over 10 years, Cobb-Douglas production with $\beta^m = 0.30$, AR(1) Hicks-neutral productivity, logit labour supply and plants competing Nash–Bertrand in wages. Two data-generating processes: one in which the labour coefficient is the same for every plant (Hicks-neutral), and one in which it is drawn $U[1/3, 2/3]$ (factor-biased). In both, the true markdown is the same.

Table 6 is the case for the whole exercise. Under the Hicks-neutral process both estimators work, as they should: the production-approach estimator returns a labour coefficient of 0.492 against a truth of 0.500. Under factor bias it collapses. It puts the labour coefficient at 0.085 against a truth of 0.501, loads the difference onto materials (0.935 against 0.300), and reports a *negative* inverse supply elasticity — an economically meaningless markdown. The joint estimator recovers the truth in both worlds, including

Table 6: Monte Carlo validation, reproducing the design of [Rubens et al. \(2025b\)](#): 50 markets $\times$ 5 plants $\times$ 10 years, Cobb-Douglas production with $\beta^m = 0.30$, AR(1) Hicks-neutral productivity, and Nash–Bertrand wage competition under logit labour supply. **Convention:** this design follows the IJIO paper in writing ψ for the inverse residual labour-supply elasticity itself, with markdown $\mu^w = \psi/(1 + \psi)$; the main text follows the AER: Insights paper in writing $\psi^l = 1 + \psi$. The markdown is the same object either way.

	mean β^l	sd β^l	β^m	ψ
DGP 1: Hicks-neutral ($\beta^l = 0.5$ for every plant)				
Truth	0.500	0.000	0.300	0.537
Hicks-neutral estimator	0.492	—	0.310	0.477
RWX joint estimator	0.501	0.000	0.299	0.535
DGP 2: factor-biased ($\beta^l \sim U[1/3, 2/3]$)				
Truth	0.501	0.094	0.300	0.537
Hicks-neutral estimator	0.085	—	0.935	-0.913
RWX joint estimator	0.501	0.094	0.300	0.525

Under a Hicks-neutral world both estimators work. Under factor bias the Hicks-neutral estimator collapses: it puts the labour coefficient at 0.085 against a truth of 0.501, loads the difference onto materials (0.935 against 0.300), and returns a *negative* inverse supply elasticity, i.e. an economically meaningless markdown. The joint estimator recovers the truth including the *dispersion* of β^l (0.094 against 0.094).

the dispersion of the labour coefficient (0.094 against a true 0.094), which is the object the Hicks-neutral model cannot represent at all.

This is a sharper statement than “the Hicks-neutral estimator is biased”. It is that the direction and magnitude of its error are governed by something the estimator does not model, and that when factor bias is present the error need not be small or even sign-preserving.

10.2 Identities checked at run time

Two identities must hold exactly if the algebra of Section 4.4 and its implementation are both right, and both are asserted rather than inspected:

1. $\ln \psi_{CES} = \ln \psi^l$ for every plant-year and every σ . The maximum absolute deviation over 289,760 plant-years is 4.8×10^{-15} at $\hat{\sigma}$, and remains below 1.6×10^{-14} at $\sigma = 0.4, 1.5$ and 3.0 .
2. The decomposition (19) must load on $(\tilde{a}, u)$ with coefficients exactly $(-k, +k)$ within industry. The regression returns

$$(-0.519231, +0.519231) \quad \text{extagainsattheoretical} \quad (-k, +k) = (-0.519231, +0.519231).$$

Appendix B gives the detail. We emphasise these because a paper whose headline rests on an invariance claim ought to demonstrate the invariance numerically rather than assert

it from the algebra alone.

10.3 Comparison with the original application

Where the two applications can be compared directly, they agree on the things that are properties of the method and differ on the things that are properties of the economy. [Rubens et al.](#) find a nesting parameter of -0.019 , statistically indistinguishable from zero, and conclude that industries and localities are close to symmetric substitutes for workers; we find -0.114 at the two-digit nest and reject the nest at its boundary, reaching the same specification. They report returns to scale between 0.983 and 1.046; we find a by-industry median of 1.085. They report ρ^ω near 0.98; we find 0.85 to 0.89. Their σ is 0.397 with a standard error of 0.215; ours is 2.08 with an unbounded robust set — the two are not distinguishable given either paper’s precision, which is itself the point of Section 8.

The substantive divergence is the one discussed in Section 11.4: the sign of the Hicks-neutral bias.

11 Results

Table 7: Headline estimates. Weighted statistics use the ASI multiplier and describe the population of registered factories; unweighted statistics reconcile with the estimation logs. Intervals are 2.5–97.5 percentiles of a firm-cluster bootstrap that re-runs the *entire* chain, so Stage-1 uncertainty is propagated. **The bootstrap statistic is the unweighted median**, so the interval belongs beside the unweighted column; the weighted column is the population-representative point estimate and has no interval of its own.

	Weighted	Unweighted	Bootstrap 95%
Median ψ^l (labour supply)	2.231	2.120	[1.657, 3.212]
Median wage markdown μ^w	0.552	0.528	[0.396, 0.689]
Median ψ_{CD} (Hicks-neutral, WP1)	0.879	0.825	–
Median θ^l (CES)	0.244	0.243	[0.012, 0.352]
Median θ^l (Cobb-Douglas, WP1)	0.114	0.100	–
Median markup μ^p (CES)	1.050	1.055	[0.046, 1.144]
Median markup μ^p (Cobb-Douglas, WP1)	1.195	1.209	–
Median $\ln \psi_{CD} - \ln \psi^l$	-0.976	-0.989	[-1.400, -0.733]

11.1 Does the monopsony finding survive?

Table 7 gives the answer, and it is: the conclusion survives and strengthens, but the evidence for it does not come from where working paper 1 thought it did.

The labour-supply markdown has a weighted median of $\psi^l = 2.23$, a **55% wage markdown**; unweighted, 2.12 and 53%. The Hicks-neutral estimate on the same plants has a weighted median of 0.88. Because $\psi^l < 1$ means workers are paid *above* marginal revenue

product, the Hicks-neutral estimator says the median Indian factory has no monopsony power at all: only 45.7% of factories have $\psi_{CD}^l > 1$. So the two estimators disagree not just about magnitude but about whether the phenomenon exists at the median plant.

They also disagree about *which* plants have power. The plant-level correlation between the two markdown series is **0.015** (Spearman 0.035). This is the central empirical fact of the paper. Two estimators applied to the same 289,760 plant-years, both claiming to measure the wedge between the wage and marginal revenue product, produce series that are statistically unrelated. They are not noisy measurements of a common object.

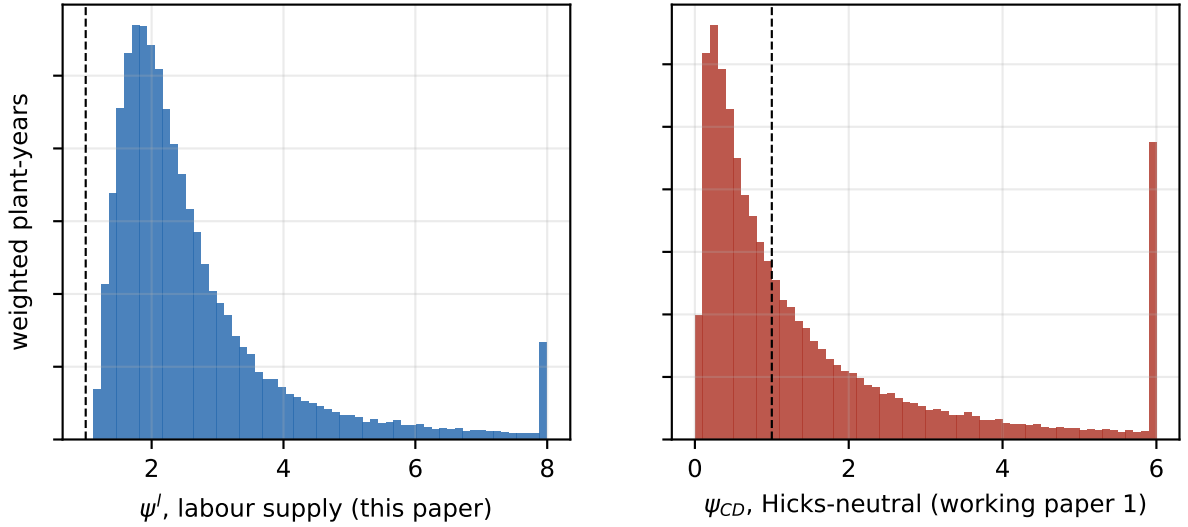


Figure 1: Weighted distributions of the two markdown measures over the same 289,760 plant-years. Left: the labour-supply estimate of this paper, which exceeds one mechanically. Right: the Hicks-neutral estimate of working paper 1, which places most of the mass *below* one. Both are weighted bin counts and therefore aggregates; the smallest non-empty bin holds 268 plant-years.

An important caution about what we may infer. Under the plain logit, $\psi^l > 1$ mechanically. The *sign* of our markdown therefore carries no information, and we do not offer “100% of plants have $\psi^l > 1$ ” as evidence of anything. What is informative is (i) the level, which the Anderson–Rubin set bounds at 33–72%, and (ii) the comparison with the Hicks-neutral estimate, which is what the next subsection is about.

11.2 How large is this, in aggregate?

Median markdowns are informative about the typical plant, but the economically relevant question is how much of the wage bill sits in plants with substantial wage-setting power. Two summary numbers, both weighted:

The **employment-weighted** median markdown is 0.461, against a plant-weighted 0.552. Larger plants mark wages down *less*. That is the direction the model predicts —

ψ^l falls with the wage and larger plants pay more — and it means the plant-weighted figure overstates the share of the aggregate wage bill that is affected. Second, plants with above-median markdowns account for 26.8% of the wage bill. So the phenomenon is widespread across plants but concentrated among smaller ones.

The right way to read the level is through the Anderson–Rubin bound of Section 7: the median markdown lies between 33% and 72%. Even the bottom of that range implies that a third of the marginal revenue product of labour at the median registered factory is retained by the employer. The top of it would be extraordinary. We think the width of this interval, rather than the point estimate, is the honest headline of the labour-supply stage, and we have reported it that way throughout.

An implication worth stating for the misallocation literature: markdown dispersion is a source of measured allocative inefficiency, and working paper 2 of this series computed it from the Hicks-neutral markdowns. Figure 1 shows the two distributions have different shapes as well as different locations, so that calculation does not survive unchanged. Redoing it is beyond this paper’s scope but it follows directly from these estimates.

11.3 The decomposition: where the Hicks-neutral estimator goes wrong

Both estimators evaluate the same identity (3). The entire difference between them is therefore the elasticity ratio:

$$\ln \psi_{CD} - \ln \psi^l = \ln \frac{\theta_{CD}^l}{\theta_{CD}^m} - \ln \frac{\theta_{CES}^l}{\theta_{CES}^m}. \quad (19)$$

This holds to 4.9×10^{-15} in our data. The Cobb–Douglas ratio is a constant within an industry by construction; the CES ratio is x_{ft} , which moves with labour-augmenting productivity. The dispersion is the story: the CES labour elasticity has a standard deviation of 0.234 against 0.041 for the Cobb–Douglas one, a factor of 5.7.

Substituting (16), the gap must load on $(\tilde{a}, u)$ with coefficients exactly $(-k, +k)$ once industry effects absorb the Cobb–Douglas constant. Regressing it returns

$$(-0.519231, +0.519231) \quad \text{against } (-k, +k) = (-0.519231, +0.519231),$$

a maximum deviation of 2.1×10^{-15} . **The discrepancy between the Hicks-neutral markdown and the true one *is* labour-augmenting productivity**, precisely as Rubens et al.’s identification result predicts, and here it is not an approximation but an identity recovered numerically.

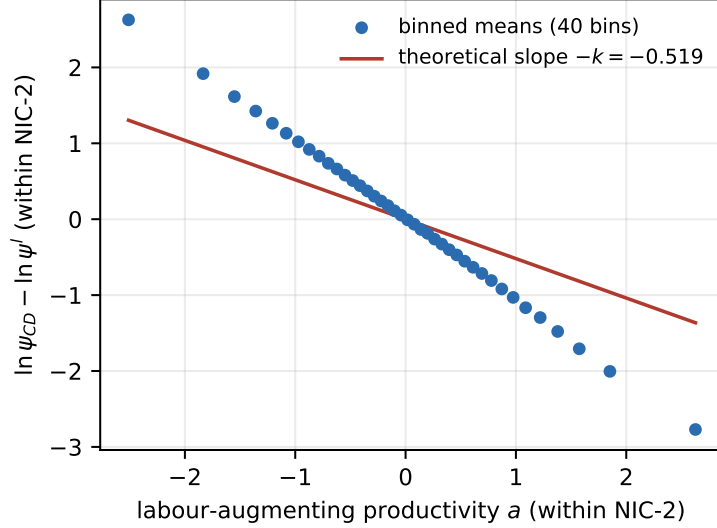


Figure 2: The decomposition, drawn. Binned means of the log-gap between the two markdown measures against labour-augmenting productivity, both taken as deviations from two-digit industry means so that the Cobb-Douglas constant is removed. The line is not fitted: it is the theoretical slope $-k = (1 - \sigma)/\sigma$ implied by (16). Forty bins; the smallest holds 7,239 plant-years.

11.4 The reversal

The median log-gap is -0.99 : the Hicks-neutral markdown is a factor **0.37** of the true one. Rubens et al. find the opposite in Chinese nonferrous metals, where the Hicks-neutral estimator reports a median markdown of 55% against a true 27% — roughly double — and manufactures a doubling trend (35% to 73%) where the true series is flat near 27%.

The sign of the bias is not a property of the estimator. By (19) it is determined by whether the plants with high labour-augmenting productivity are the ones with high or low labour cost shares. In Chinese NFM mining and smelting over 1999–2006, technology was moving sharply in a labour-saving direction — deep-hole drilling replacing shrinkage stope mining — and labour-augmenting productivity grew at 15.1% a year; the Hicks-neutral estimator read that directed technical change as rising market power. Indian registered manufacturing over 2012–2022 is a much more heterogeneous population in which the sign of that covariance goes the other way. The implication for practice is uncomfortable but clear: **one cannot sign the bias of a Hicks-neutral markdown estimate *a priori***. It is not conservative in either direction. It has to be measured, which requires exactly the outside restriction Rubens et al. propose.

11.5 Elasticities and markups

The labour elasticity roughly doubles: a weighted median of 0.244 under CES with labour augmentation against 0.114 under Cobb-Douglas. Correspondingly the materials elasticity falls and the estimated markup falls, from a weighted median of 1.195 to 1.050. **Allowing**

Table 8: By year. ψ^l is the labour-supply markdown of this paper, ψ_{CD} the Hicks-neutral estimate of working paper 1. **The decline in ψ^l is largely mechanical:** γ is estimated once for the whole window while real wages rise, and $\psi = 1 + 1/(\gamma W(1 - S))$ falls in W by construction. It is *not* evidence that monopsony power fell. The log-gap column does not share this mechanism.

Year	ψ^l	μ^w (%)	ψ_{CD}	θ^l	μ^p	$\ln \psi_{CD} - \ln \psi^l$	N
2013	2.743	63.5	0.938	0.266	1.013	-1.106	26,582
2014	2.621	61.9	0.916	0.262	1.021	-1.078	27,917
2015	2.482	59.7	0.858	0.270	1.026	-1.108	26,507
2016	2.353	57.5	0.822	0.261	1.037	-1.086	27,638
2017	2.250	55.6	0.831	0.255	1.047	-1.045	28,487
2018	2.169	53.9	0.892	0.238	1.054	-0.929	29,156
2019	2.096	52.3	0.811	0.244	1.056	-0.985	29,616
2020	2.105	52.5	0.820	0.244	1.063	-0.962	30,379
2021	1.998	49.9	0.943	0.210	1.071	-0.778	31,131
2022	1.937	48.4	0.968	0.202	1.079	-0.733	32,347

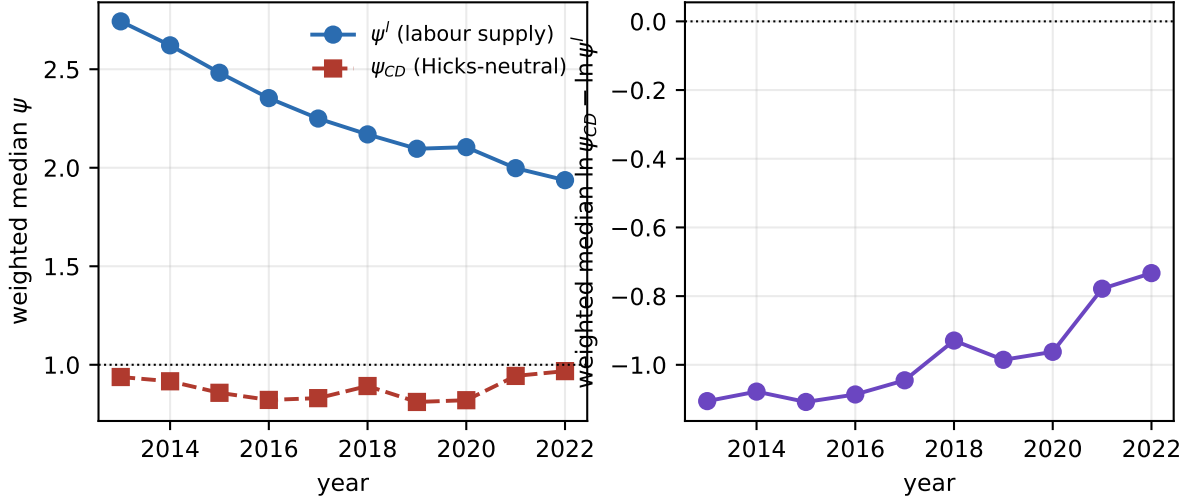


Figure 3: Weighted medians by year. Left: the two markdown measures. **The decline in ψ^l is mechanical** — γ is constant while real wages rise — and is not evidence about market power; the point of the panel is that the Hicks-neutral series does not track it. Right: the log-gap, which is not a wage transform, and which narrows over the decade.

labour-augmenting technology moves measured product market power down and measured labour market power up. That is a coherent story: the Hicks-neutral model was attributing to the product market a wedge that belongs to the labour market.

11.6 Labour-augmenting productivity

Recovering A_{ft} was the object of the exercise as much as the markdown was, and it carries two restrictions that we state before any number.

Its level is a normalisation. Section 4.4 shows only $b = \beta^k/\beta^m$ is identified; the scale of A is absorbed into β^m by construction. Levels are not comparable to anything outside this paper.

Its aggregate time path is not a technology trend. a is built from $v = w^l + \ln \psi^l - w^m$, and w^m is the common wholesale price index used to deflate materials, so the year-by-year common component of a is identified only *relative to that index*. The year effects in estimation absorb it — which is why σ is unaffected — but it means that a headline “labour-augmenting productivity growth rate” computed from our a would be a price index in disguise. We therefore report none, and we note that this is a real difference from Rubens et al. (2025a), who report 15.1% annual labour-augmenting productivity growth in Chinese nonferrous metals. Their design has firm-level output and input prices; ours does not.

What *is* identified is the cross-section within a year, because the normalisation and the price index are common to all plants in that year. Labour-augmenting productivity is widely dispersed: the within-year weighted standard deviation of a is 1.151 log points and the p90–p10 spread is 2.870, both medians across years. For comparison, the entire Cobb-Douglas labour elasticity in working paper 1 has a standard deviation of 0.041. **A model that forces this dispersion into a single industry constant has to put it somewhere, and the production approach puts it into the markdown** — which is Figure 2 restated.

The cross-sectional gradients are interpretable and they are economically sensible. Government plants have the highest a of any ownership category (Table 9), which is consistent with their being larger, more capital-intensive and more formal. Exporters have higher a than non-exporters. Both patterns are independent of the wage mechanism that Table 12 warns about, because a enters through the input ratio rather than through the wage level alone.

11.7 Ownership

Government plants have the lowest markdown of any ownership category — $\psi^l = 1.60$, a 37% markdown, against 2.62 and 62% for proprietary and partnership firms — and the highest labour-augmenting productivity. The ordering is the same under both estimators,

Table 9: Ownership. The 2017–2022 window is the clean one: the ASI ownership code list breaks in 2015–16 (Appendix D). The wage column is reported because the ranking of ψ^l across any cross-section is, under a single γ , the inverse wage ranking almost by construction; see Table 12.

	ψ^l	μ^w (%)	ψ_{CD}	θ^l	μ^p	Wage (Rs)	N
<i>2017-2022 (clean B02 codes)</i>							
Proprietary / partnership	2.618	61.8	0.901	0.270	1.023	139,176	63,214
Other	2.312	56.8	0.292	0.472	0.810	171,552	892
Co-operative	1.879	46.8	0.921	0.174	1.019	256,134	3,662
Private corporate	1.827	45.3	0.868	0.207	1.095	272,382	110,916
Government	1.598	37.4	0.621	0.269	1.021	376,739	2,432
<i>full window 2013-2022</i>							
Proprietary / partnership	2.856	65.0	0.893	0.292	1.001	121,287	102,355
Other	2.535	60.6	0.326	0.480	0.815	146,643	1,593
Co-operative	1.960	49.0	0.842	0.194	1.002	234,548	6,055
Private corporate	1.923	48.0	0.881	0.213	1.086	244,011	175,713
Government	1.648	39.3	0.604	0.279	1.008	347,676	4,044

which is a useful external check. It is also the pattern public-sector pay rules predict, and it agrees in direction with Rubens et al., who find markdowns 13% lower at Chinese state-owned enterprises.

We must immediately qualify it. Table 12 shows that the ownership ranking of ψ^l is the inverse ownership ranking of wages, at a rank correlation of -1.000 . Government plants pay more; under a single γ that is the finding about markdowns. The independent content lies in the ψ_{CD} column, which is not a wage transform and which also ranks government lowest, and in the labour-augmenting productivity column, which is where the substantive difference between public and private plants shows up.

11.8 States and industries

Tables 10 and 11 report the geographic and industrial distribution, and Table 12 is the instruction for how to read them. The rank correlation between a group’s median ψ^l and its median wage is -1.000 for states, for industries and for ownership types. The apparent finding that markdowns are highest in Tripura, Bihar, Mizoram, Nagaland and Manipur, and lowest in Goa and Maharashtra, is **the statement that wages are lowest in the former and highest in the latter**, which we already knew. We do not present it as a result.

The log-gap column does carry independent information: its rank correlation with the wage is 0.42 across states and 0.05 across industries. Where it is most negative — Mizoram, Tripura and Bihar, and the printing, textiles and apparel divisions — the Hicks-neutral estimator understates the markdown most severely, and by (19) those are the places where labour-augmenting productivity is furthest from its industry norm.

Table 10: By state, ordered by the estimated markdown. Read with Table 12: the ψ^l ordering is the inverse wage ordering by construction, so the informative column is the log-gap against the Hicks-neutral estimate, which is not a transformation of the wage.

State	ψ^l	μ^w (%)	ψ_{CD}	$\ln \psi_{CD} - \ln \psi^l$	Wage (Rs)	N
Tripura	5.532	81.9	0.449	-2.591	49,683	2,930
Bihar	5.179	80.7	0.354	-2.540	53,873	6,755
Mizoram	4.919	79.7	0.213	-3.125	57,460	358
Nagaland	4.679	78.6	0.424	-2.372	61,200	1,003
Manipur	4.513	77.8	0.476	-2.245	64,098	1,366
Assam	3.759	73.4	0.825	-1.449	81,608	7,361
Arunachal Pradesh	3.277	69.5	0.668	-1.666	98,886	653
Jharkhand	2.686	62.8	1.139	-0.823	133,514	3,660
Andhra Pradesh	2.680	62.7	1.197	-0.888	134,004	8,843
Odisha	2.680	62.7	1.141	-0.856	134,011	4,928
Meghalaya	2.656	62.3	1.509	-0.473	136,004	1,010
Chhattisgarh	2.643	62.2	1.574	-0.518	137,058	4,493
Jammu & Kashmir	2.608	61.7	1.382	-0.678	140,059	2,803
Punjab	2.537	60.6	0.905	-1.041	146,485	11,727
West Bengal	2.419	58.7	1.057	-0.898	158,633	12,187
Madhya Pradesh	2.343	57.3	1.269	-0.666	167,630	10,645
Tamil Nadu	2.310	56.7	0.607	-1.354	171,913	36,213
Uttar Pradesh	2.308	56.7	0.991	-0.929	172,204	26,738
Kerala	2.240	55.4	0.711	-1.285	181,639	8,504
Uttarakhand	2.210	54.7	1.228	-0.613	186,138	8,082
Rajasthan	2.184	54.2	1.073	-0.733	190,111	13,344
Telangana	2.139	53.3	1.059	-0.783	197,673	7,973
Chandigarh	2.133	53.1	0.885	-1.044	198,671	662
Puducherry	2.094	52.2	0.984	-0.800	205,901	1,793
Andaman & Nicobar	2.092	52.2	0.504	-1.468	206,278	81
Gujarat	2.076	51.8	1.079	-0.689	209,320	25,529
Haryana	1.995	49.9	0.777	-0.953	226,355	16,708
Delhi	1.980	49.5	0.682	-1.106	229,725	4,915
Karnataka	1.954	48.8	0.735	-1.010	236,079	16,802
Himachal Pradesh	1.935	48.3	1.287	-0.429	240,793	5,489
Sikkim	1.931	48.2	1.133	-0.591	241,956	586
Maharashtra	1.849	45.9	0.759	-0.943	265,181	33,554
Goa	1.716	41.7	0.683	-0.926	314,414	2,062

Table 11: By NIC-2 division. ν is that division’s own returns-to-scale estimate from Stage 3. Division 33 has no Cobb-Douglas counterpart in working paper 1, so its comparison columns are blank rather than imputed.

NIC	Division	ψ^l	μ^w (%)	ψ_{CD}	$\ln \psi_{CD} - \ln \psi^l$	θ^l	ν	N
10	Food products	2.429	58.8	1.696	-0.379	0.137	1.085	43,021
11	Beverages	2.358	57.6	1.216	-0.709	0.255	1.150	6,834
12	Tobacco	3.309	69.8	0.644	-1.611	0.478	1.030	3,943
13	Textiles	2.284	56.2	0.345	-1.893	0.229	1.045	21,369
14	Wearing apparel	2.255	55.7	0.366	-1.811	0.342	0.980	13,914
15	Leather	2.267	55.9	0.578	-1.371	0.293	1.005	6,264
16	Wood	3.034	67.0	1.333	-0.821	0.282	1.125	5,612
17	Paper	2.416	58.6	1.369	-0.601	0.205	1.070	7,468
18	Printing/recorded media	1.996	49.9	0.213	-2.254	0.295	0.955	4,505
19	Coke & refined petroleum	2.069	51.7	1.745	-0.308	0.085	0.995	3,524
20	Chemicals	2.066	51.6	1.361	-0.464	0.206	1.140	19,329
21	Pharmaceuticals	1.785	44.0	1.141	-0.482	0.321	1.160	11,155
22	Rubber & plastics	2.167	53.9	1.263	-0.598	0.203	1.075	13,935
23	Non-metallic minerals	3.319	69.9	0.724	-1.461	0.535	1.205	24,491
24	Basic metals	2.182	54.2	1.253	-0.568	0.101	1.035	17,064
25	Fabricated metal	2.237	55.3	0.846	-1.025	0.275	1.090	14,703
26	Computer/electronic/optical	1.852	46.0	0.551	-1.248	0.336	1.105	6,162
27	Electrical equipment	2.020	50.5	1.085	-0.681	0.209	1.065	16,101
28	Machinery n.e.c.	1.925	48.1	0.525	-1.315	0.308	1.090	18,093
29	Motor vehicles	1.853	46.0	0.603	-1.143	0.246	1.095	14,263
30	Other transport	2.030	50.7	0.712	-1.099	0.250	1.085	5,960
31	Furniture	2.377	57.9	1.136	-0.800	0.267	1.100	3,910
32	Other manufacturing	2.019	50.5	0.737	-1.088	0.315	1.105	7,979
33	Repair & installation	2.448	59.2	—	—	0.456	—	161

Table 12: How much of each cross-sectional gradient is mechanical. Under the plain logit with a single γ , ψ^l is a decreasing function of the plant’s own wage, so any ranking of ψ^l reproduces the inverse wage ranking. Spearman rank correlations of group medians. The log-gap against the Hicks-neutral estimate does not inherit the mechanism.

Split	Groups	rank corr(ψ^l , wage)	rank corr(log-gap, wage)
State	33	-1.000	0.423
NIC-2 industry	24	-1.000	0.046
Ownership (2017–22)	5	-1.000	—

Exporters have lower markdowns than non-exporters (1.96 against 2.28), again with the caveat that exporters pay more.

12 Robustness and what is weakly identified

12.1 The bootstrap

The stage-by-stage standard errors treat ψ^l as data when it is an estimate. We therefore resample *plants* with replacement and re-run the entire chain — $\gamma \rightarrow \psi^l \rightarrow (\sigma, \rho^a) \rightarrow (b, \nu, \rho^\omega) \rightarrow$ elasticities, markups, markdowns and the gap — so that Stage-1 uncertainty propagates into everything downstream. A resampled plant enters as a distinct cluster. Intervals are reported in Table 7. The markdown line is the one to rely on; the σ line is wide, and a bootstrap reproduces weak identification rather than curing it.

12.2 Sensitivity to σ

Table 13: Sensitivity to σ , which is only weakly identified. The final column is the point of the exercise: the markdown is *identical* in every row, because the first-order condition $\ln(\theta^l/\theta^m) = \ln(\alpha^l\psi^l/\alpha^m)$ holds identically for every σ under the normalisation of Section 4.4. Only the production-side objects move.

σ	ν	θ^l	θ^m	θ^k	μ^p	$\text{sd}(a)$	ψ^l
0.40	0.510	0.1195	0.3889	0.0001	0.516	2.241	2.120
0.80	0.795	0.1869	0.6080	0.0001	0.807	6.510	2.120
1.50	1.170	0.2603	0.8462	0.0543	1.132	2.516	2.120
2.08	1.110	0.2505	0.8146	0.0345	1.092	1.160	2.120
3.00	1.085	0.2467	0.8025	0.0248	1.076	0.652	2.120
2.08	1.090	0.2420	0.7864	0.0474	1.058	1.160	2.120

Table 13 is the practical payoff of Section 4.4. Across σ from 0.40 to 3.00 — a range that spans Rubens et al.’s Chinese estimate, the Cobb-Douglas benchmark and our own point estimate — the elasticities, the markup and the dispersion of A all move, and the markdown **does not move at all**. It is identical to displayed precision in every row. Readers who find $\hat{\sigma} = 2.08$ implausibly high (it implies labour and materials are gross substitutes, where Rubens et al. find gross complements) may substitute their preferred value and the paper’s headline is unchanged.

12.3 Robustness already reported elsewhere

Three checks were run at earlier stages and we summarise them rather than repeat them. The 2019 wave reports 3.9% exporters against 7.1–8.4% elsewhere, and through the lag this contaminates the 2020 instrument; dropping 2020 gives $\gamma = 0.430$ against 0.444. Dropping

the carried-forward minimum-wage observations leaves σ within its (unbounded) robust set. Adding a minimum-wage “bite” interaction — the plant’s lagged distance above the statutory floor, interacted with the change in the floor, which is the standard Indian design following Soundararajan (2019) — leaves $\sigma = 2.085$ and improves the Hansen p to 0.662 without narrowing the robust set.

12.4 Limitations we cannot repair

1. **Weak identification of σ** , diagnosed in Section 8. Mitigated, not solved, by the invariance result.
2. **No time-varying γ_t** . Rubens et al. estimate one and find it declining. Our instruments cannot support it, so our time series and cross-sections of ψ^l are mechanically wage-driven and are not interpreted.
3. **States, not commuting zones**. No district identifier exists in the released ASI records.
4. **No plant-level output price**, so the De Loecker et al. (2016) critique applies to ω and hence to ν and the markup, though not to the markdown.
5. **Minimum wages end in 2020**. Stage 2 uses 2013–2020; Stages 1 and 3 use the full window.
6. **Revenue-based shares**. α^l and α^m are computed from revenue, as in working paper 1, so that the two papers are comparable.

13 Implications

For this research series. Working papers 1 and 4 cannot both be read at face value, and this paper says which part of each survives. Working paper 1’s *conclusion* — that Indian manufacturing workers are paid materially below marginal revenue product — is supported and strengthened: the markdown is larger under the labour-supply estimator than under the Hicks-neutral one. Working paper 1’s *cross-section* does not survive: its plant-level ranking of who exercises labour market power is uncorrelated with the ranking obtained once labour-augmenting technology is admitted. Any downstream use of plant-level Hicks-neutral markdowns — including the misallocation decompositions of working paper 2 — inherits that problem and should be revisited.

For the markdown literature. The practical lesson is the reversal of Section 11.4. It is tempting to treat Rubens et al.’s Chinese result as establishing that Hicks-neutral estimators *overstate* markdowns, and therefore to treat existing estimates as upper bounds.

Our result shows the bias has no fixed sign: it depends on a covariance that differs across settings. A Hicks-neutral markdown estimate is not conservative in either direction.

For minimum wage policy in India. If the median registered factory retains roughly half of the marginal revenue product of labour — with an Anderson–Rubin bound of 33–72% — then the textbook monopsony case for a binding wage floor raising both wages and employment applies over a wide range. We stop short of quantifying it: doing so requires the time-varying γ_t that our instruments cannot support, and the minimum-wage series we assembled here is an input to identification rather than the object of a policy evaluation. That evaluation is the natural next paper, and the series is now available for it.

14 Conclusion

Rubens et al. (2025a) show that wage markdowns and labour-augmenting productivity are not separately identified from production data alone. We implement their remedy on Indian registered manufacturing and find that the median plant retains about half of labour’s marginal revenue product, that the Hicks-neutral estimate of the same object is a factor 0.37 of it, and that the two are uncorrelated across plants. We prove that under the natural CES normalisation the markdown is invariant to the production parameters, verify the resulting identity to machine precision, and show that the entire discrepancy with the Hicks-neutral estimator loads on labour-augmenting productivity with a coefficient we recover exactly. And we find that the direction of the bias reverses the Chinese benchmark, which means the bias cannot be signed without doing the work.

We have been explicit about what the Indian data cannot support: the elasticity of substitution is weakly identified, the cross-sectional patterns in ψ^l are mechanically wage-driven under a constant wage coefficient, and labour markets are defined at the state rather than the commuting zone. The first of these does not touch the headline, and we show why; the second and third bound what may be inferred from the distributional tables, and we say so where the tables appear rather than in a footnote.

A Derivation of the input-ratio equation

From (8), $\theta^l/\theta^m = (AL)^k/(\beta^m M^k)$. Setting this equal to the first-order condition (3) value $\alpha^l\psi^l/\alpha^m$ and taking logs with $x = \ln(\alpha^l\psi^l/\alpha^m)$:

$$k(a + l) - \ln \beta^m - k m = x.$$

Solving for a and multiplying by $(1 - \sigma)$, using $(1 - \sigma)/k = -\sigma$:

$$(1 - \sigma)a = -\sigma(x + \ln \beta^m) - (1 - \sigma)(l - m) = -\sigma x + (1 - \sigma)(m - l) - \sigma \ln \beta^m,$$

which with $u = m - l$ and $v = u + x$ rearranges to (11), the constant being $c = \sigma \ln \beta^m$. Substituting into the AR(1) of (9) gives the bilinear residual (12); the constant enters as $(1 - \rho^a)c$ and is absorbed by the intercept and fixed effects.

To see that v is the markdown-adjusted relative price, note $x = \ln(W^l L \psi^l) - \ln(W^m M)$, so $v = u + x = (m - l) + (w^l + l + \ln \psi^l) - (w^m + m) = w^l + \ln \psi^l - w^m$.

B Unit tests

Two identities must hold exactly if the algebra and the code are right. Both are asserted in the estimation script rather than checked by eye.

1. **The first-order condition, for every σ .** Over all 289,760 plant-years, $\max |\ln \psi_{CES} - \ln \psi^l|$ equals 4.8×10^{-15} at $\hat{\sigma}$, and 1.6×10^{-14} , 4.6×10^{-15} , 6.5×10^{-15} at $\sigma = 0.4, 1.5, 3.0$ respectively.
2. **The decomposition loading.** Regressing $\ln \psi_{CD} - \ln \psi^l$ on $(\tilde{a}, u)$ within two-digit industry returns $(-0.519231, +0.519231)$ against the theoretical $(-k, +k)$, a maximum deviation of 2.1×10^{-15} .

The estimator was additionally validated on the Monte Carlo design of [Rubens et al. \(2025b\)](#): under a Hicks-neutral data-generating process the plain production-approach markdown is consistent, and under a factor-biased one it is severely biased while the joint estimator recovers the truth.

C The minimum wage series

The applied literature on Indian minimum wages traces to a single office. [Mansoor and O'Neill \(2021\)](#) assemble more than 1,500 Indian minimum wages and are the reference dataset, collated from the Labour Bureau; [Menon and Rodgers \(2017\)](#) build their own from state rates; [Soundararajan \(2019\)](#) uses state-time variation in the same statutory rates. [Mansoor and O'Neill](#)'s series covers 1999–2011 and therefore ends before our window begins, so we go to the primary source.

Source. Labour Bureau, Ministry of Labour & Employment, Government of India, *Report on the Working of the Minimum Wages Act, 1948* (annual), Table 4, “Range of Minimum Wages as on 31.12.year”, which reports for each State and Union Territory the number of scheduled employments and the lowest and highest minimum wage in rupees

per day for unskilled workers. Nine annual reports (2012–2020) were downloaded and parsed.

Validation. Each candidate row must satisfy the report’s own arithmetic, $\text{range} = \text{highest} - \text{lowest}$, to within 0.05; rows failing are discarded and counted, so a misaligned parse cannot pass silently. The result is 321 state-years across 36 states. The only unmapped label is “C.L.C. (Central)”, the Chief Labour Commissioner’s central sphere, which is correctly not a state. As an independent check, the parsed Delhi 2020 rate of Rs 596.00 matches the narrative text of the 2020 report.

Footnote markers are recorded, not discarded. An asterisk means no annual return was received and the figures are repeated from the previous report; other markers denote partial repetition or a “plus subsidised food” qualifier on the value. Sixty-three state-years (19.6%) are carried forward and are flagged so they can be dropped: they carry no independent time variation and would otherwise manufacture spurious “no change” observations. This also explains apparent jumps — Gujarat’s move from 100 to 268 between 2015 and 2016 is because the 2015 figure is stale, not because 2016 moved.

Gaps, reported not filled. No report exists for 2021 or 2022, so those panel years have no minimum wage. Merge coverage on the Stage-1 sample is 77.6%, of which 18.7% are carried forward, leaving 58.9% independently reported. India sets minimum wages by scheduled employment and most states do not vary them by manufacturing industry, so $\text{state} \times \text{year}$ is the correct level; a $\text{state} \times \text{industry}$ series would be largely fictitious.

D Ownership recovery from the raw records

Two traps in the raw ASI blocks would silently corrupt an ownership analysis, and both were found only by inspecting raw bytes against the official code lists.

First, **field B03 is “type of ownership” only in 2012–13 to 2014–15.** From 2015–16 the same slot holds the 21-character Corporate Identification Number, masked with nines. Read as a numeric ownership code it returns 10^{21} for three-quarters of rows. Our parser tests the declared field width and treats B03 as ownership only where that width is one character.

Second, **the B02 code list breaks in 2015–16.** Cross-tabulating B02 against B03 in the overlap period reveals that B02 is *legal form* while B03 is *ownership*: a wholly government-owned firm registered as a public limited company appears under B02= 4. A government-share indicator built from B02 alone therefore understates public ownership before 2015. We use $B03 \in \{1, 2, 3\}$ for 2012–2014 and $B02 \in \{4, 5\}$ for 2015–2022, and report 2017–2022 as the clean baseline window in Table 9.

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